Technology Stack

Date	02 October 2026
Team ID	SWTID-2026-1529
Project Name	EduGenie: Google gemini powered learning assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*[Specify technology]*	*[Explain why this was chosen (e.g., responsive design, component reusability)]*

1 Frontend / Client-Side Next.js 15, React 19, HTML, Tailwind CSS Client-side rendering, SSR, Native Audio TTS, and dynamic React UI.

2	**Backend / Server-Side**	*[Specify technology]*	*[Explain why this was chosen (e.g., fast processing, existing team expertise)]*
2	Backend / Server-Side	FastAPI, Python 3.14, Uvicorn	Fast execution, robust Pydantic JSON validation.
3	Database / Storage	Supabase (PostgreSQL)	Seamless authentication and rapid real-time queries.
4	Cloud / Hosting / Deployment	Vercel (Frontend), Render (Backend)	Streamlined CI/CD and cost-effective scaling.
3	**Database / Data Storage**	*[Specify technology]*	*[Explain why this was chosen (e.g., handles unstructured data, strong relational mapping)]*
	(e.g., MySQL, MongoDB, PostgreSQL)		
4	**Cloud / Hosting / Deployment**	*[Specify technology]*	*[Explain why this was chosen (e.g., easy scalability, cost-effective hosting)]*
	(e.g., AWS, Azure, Heroku, Docker)		
5	**Version Control & CI/CD**	*[Specify technology]*	*[Explain why this was chosen (e.g., seamless collaboration, automated testing)]*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools**	*[Specify technology]*	*[Explain the specific feature this enables in your project]*
5	Third-Party APIs / AI	Google Gemini API (gemini-3.5-flash-lite)	Core intelligence engine for NLP & JSON MCQ generation.
	(e.g., Stripe, Twilio, Google Maps)		